

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import KNNImputer
from scipy.stats import zscore
```

Step 2: Load Dataset

```
df = pd.read_csv("movies.csv")
```

```
df.head()
```

		title_x	imdb_id	\
0		Uri: The Surgical Strike	tt8291224	
1		Battalion 609	tt9472208	
2	The Accidental Prime Minister (film)		tt6986710	
3		Why Cheat India	tt8108208	
4		Evening Shadows	tt6028796	

		poster_path	\
0		https://upload.wikimedia.org/wikipedia/en/thumb/Uri:...	
1		NaN	
2		https://upload.wikimedia.org/wikipedia/en/thumb/...	
3		https://upload.wikimedia.org/wikipedia/en/thumb/...	
4		NaN	

		wiki_link	\
0		https://en.wikipedia.org/wiki/Uri:_The_Surgica...	
1		https://en.wikipedia.org/wiki/Battalion_609	
2		https://en.wikipedia.org/wiki/The_Accidental_P...	
3		https://en.wikipedia.org/wiki/Why_Cheat_India	
4		https://en.wikipedia.org/wiki/Evening_Shadows	

		title_y	original_title
is_adult	\		
0		Uri: The Surgical Strike	Uri: The Surgical Strike
0			
1		Battalion 609	Battalion 609
0			
2		The Accidental Prime Minister	The Accidental Prime Minister
0			
3		Why Cheat India	Why Cheat India
0			
4		Evening Shadows	Evening Shadows
0			

year_of_release	runtime	genres	imdb_rating	imdb_votes
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\					
0	2019	138	Action Drama War	8.4	35112
1	2019	131	War	4.1	73
2	2019	112	Biography Drama	6.1	5549
3	2019	121	Crime Drama	6.0	1891
4	2018	102	Drama	7.3	280

	story \
0	Divided over five chapters the film chronicle...
1	The story revolves around a cricket match betw...
2	Based on the memoir by Indian policy analyst S...
3	The movie focuses on existing malpractices in ...
4	While gay rights and marriage equality has bee...

	summary	tagline \
0	Indian army special forces execute a covert op...	NaN
1	The story of Battalion 609 revolves around a c...	NaN
2	Explores Manmohan Singh's tenure as the Prime ...	NaN
3	The movie focuses on existing malpractices in ...	NaN
4	Under the 'Evening Shadows' truth often plays...	NaN

	actors
wins_nominations \	
0	Vicky Kaushal Paresh Rawal Mohit Raina Yami Ga...
4 wins	
1	Vicky Ahuja Shoaib Ibrahim Shrikant Kamat Elen...
NaN	
2	Anupam Kher Akshaye Khanna Aahana Kumra Atul S...
NaN	
3	Emraan Hashmi Shreya Dhanwanthary Snighdadeep ...
NaN	
4	Mona Ambegaonkar Ananth Narayan Mahadevan Deva... 17 wins & 1 nomination

	release_date
0	11 January 2019 (USA)
1	11 January 2019 (India)
2	11 January 2019 (USA)
3	18 January 2019 (USA)
4	11 January 2019 (India)

1. Basic Information

```
df.info()
df.describe()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1629 entries, 0 to 1628
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title_x                1629 non-null   object
1   imdb_id                1629 non-null   object
2   poster_path            1526 non-null   object
3   wiki_link              1629 non-null   object
4   title_y                1629 non-null   object
5   original_title         1629 non-null   object
6   is_adult               1629 non-null   int64
7   year_of_release        1629 non-null   int64
8   runtime                1629 non-null   object
9   genres                 1629 non-null   object
10  imdb_rating            1629 non-null   float64
11  imdb_votes             1629 non-null   int64
12  story                  1609 non-null   object
13  summary                1629 non-null   object
14  tagline                 557 non-null    object
15  actors                  1624 non-null   object
16  wins_nominations        707 non-null    object
17  release_date            1522 non-null   object
dtypes: float64(1), int64(3), object(14)
memory usage: 229.2+ KB

```

	is_adult	year_of_release	imdb_rating	imdb_votes
count	1629.0	1629.000000	1629.000000	1629.000000
mean	0.0	2010.263966	5.557459	5384.263352
std	0.0	5.381542	1.567609	14552.103231
min	0.0	2001.000000	0.000000	0.000000
25%	0.0	2005.000000	4.400000	233.000000
50%	0.0	2011.000000	5.600000	1000.000000
75%	0.0	2015.000000	6.800000	4287.000000
max	0.0	2019.000000	9.400000	310481.000000

2. Missing Value Analysis

```
df.isnull().sum()
```

```

title_x                0
imdb_id                0
poster_path            103
wiki_link              0
title_y                0
original_title         0
is_adult               0
year_of_release        0
runtime                0
genres                 0

```

```
imdb_rating      0
imdb_votes       0
story            20
summary          0
tagline          1072
actors           5
wins_nominations 922
release_date     107
dtype: int64
```

3. Missing Value Treatment

Categorical Variables

```
df['poster_path'].fillna('Unknown', inplace=True)
df['story'].fillna('No Story Available', inplace=True)
df['tagline'].fillna('No Tagline', inplace=True)
df['actors'].fillna('Unknown', inplace=True)
df['wins_nominations'].fillna('0 Wins', inplace=True)
df['release_date'].fillna(method='ffill', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['poster_path'].fillna('Unknown', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:3:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['story'].fillna('No Story Available', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:4:
FutureWarning: A value is trying to be set on a copy of a DataFrame or

Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['tagline'].fillna('No Tagline', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:5:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['actors'].fillna('Unknown', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:6:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['wins_nominations'].fillna('0 Wins', inplace=True)
```

C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:7:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the

original object.

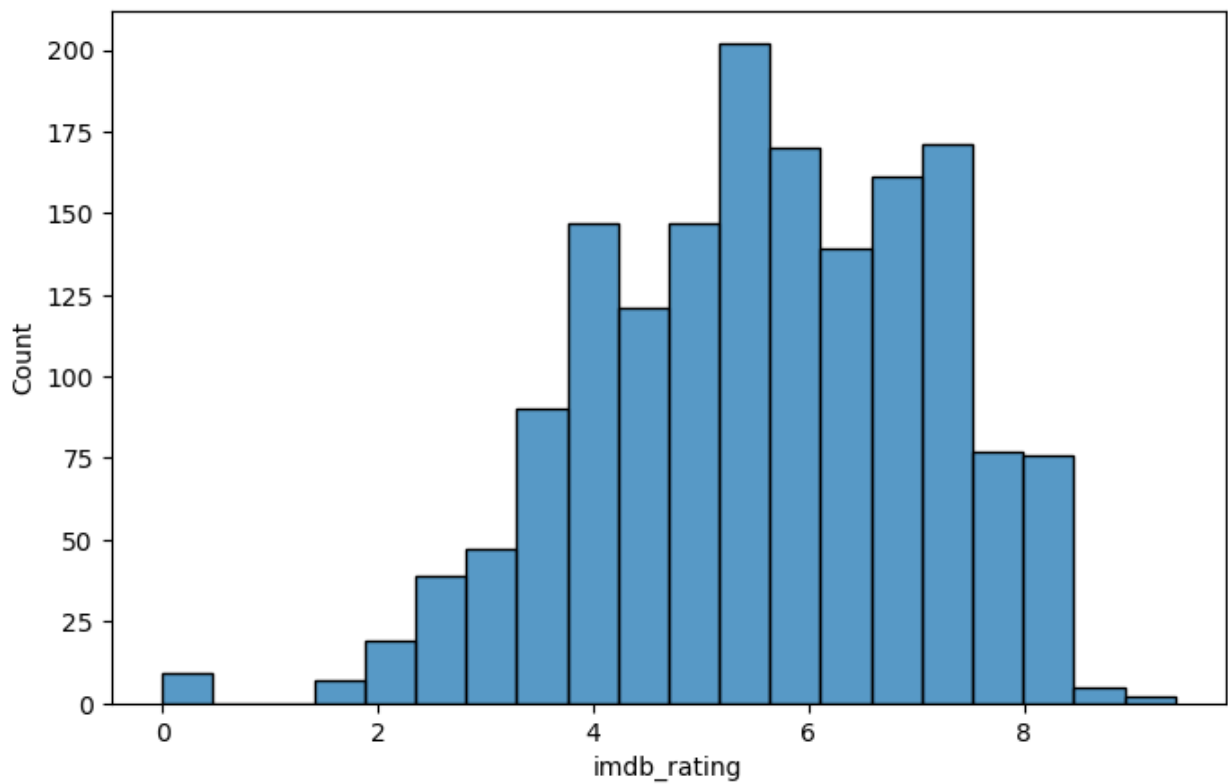
```
df['release_date'].fillna(method='ffill', inplace=True)
C:\Users\abhis\AppData\Local\Temp\ipykernel_23712\3564884867.py:7:
FutureWarning: Series.fillna with 'method' is deprecated and will
raise in a future version. Use obj.ffill() or obj.bfill() instead.
df['release_date'].fillna(method='ffill', inplace=True)
```

```
df.isnull().sum()
```

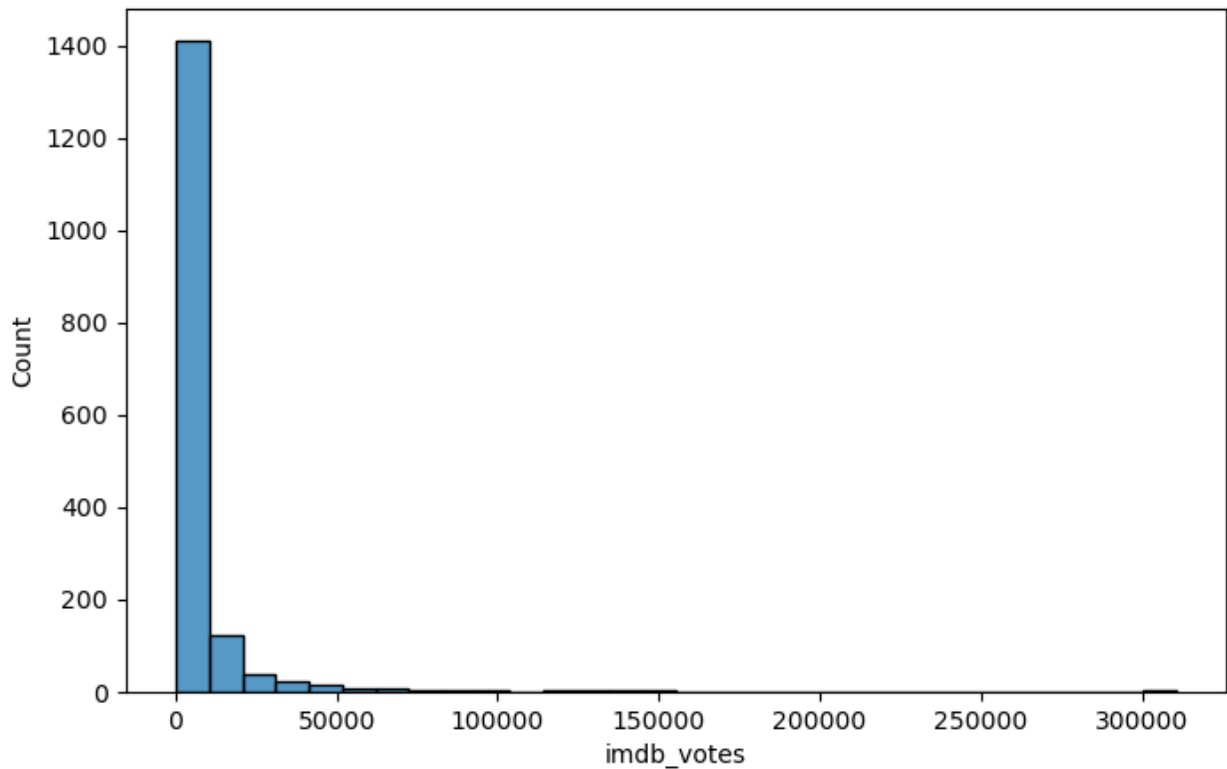
```
title_x      0
imdb_id      0
poster_path  0
wiki_link    0
title_y      0
original_title  0
is_adult     0
year_of_release  0
runtime      0
genres       0
imdb_rating  0
imdb_votes   0
story        0
summary      0
tagline      0
actors       0
wins_nominations  0
release_date  0
dtype: int64
```

4. Exploratory Data Analysis

```
## IMDb Rating Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['imdb_rating'], bins=20)
plt.show()
```



```
## IMDb Votes Distribution
plt.figure(figsize=(8,5))
sns.histplot(df['imdb_votes'], bins=30)
plt.show()
```



Top Genres

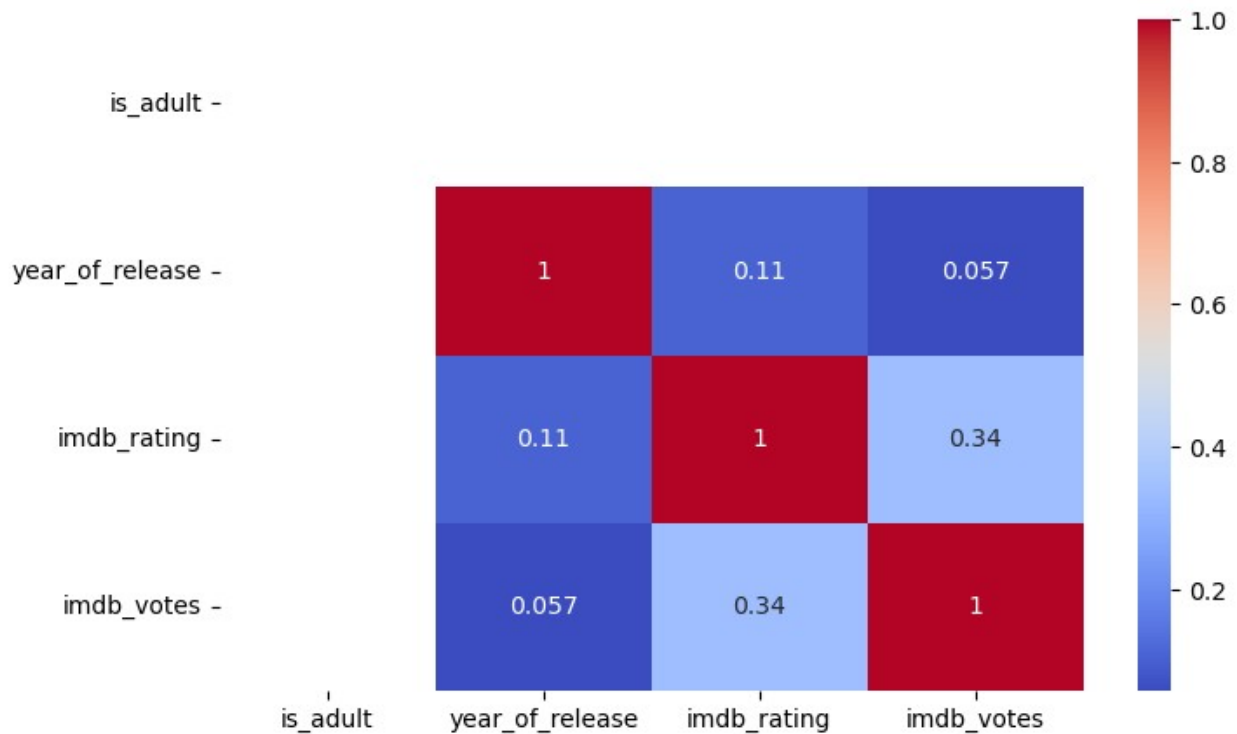
```
df['genres'].value_counts().head(10)
```

```
genres
Drama                                162
Comedy|Drama|Romance                 101
Comedy|Drama                         88
Drama|Romance                        86
Action|Crime|Drama                   86
Comedy                               73
Comedy|Romance                       69
Action|Thriller                      36
Drama|Musical|Romance                 28
Comedy|Crime|Drama                   27
Name: count, dtype: int64
```

Correlation Matrix

```
numeric = df.select_dtypes(include=np.number)
```

```
plt.figure(figsize=(8,5))
sns.heatmap(numeric.corr(),
            annot=True,
            cmap='coolwarm')
plt.show()
```

5. Outlier Detection (IQR)

```
## IMDb Votes
Q1 = df['imdb_votes'].quantile(0.25)
Q3 = df['imdb_votes'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR

## Winsorization
df['imdb_votes'] = np.clip(
    df['imdb_votes'],
    lower,
    upper
)
```

6. Outlier Detection (Z-score)

```
z = np.abs(zscore(df[['imdb_rating',
                      'imdb_votes',
                      'year_of_release'])))

outliers = np.where(z > 3)

print(outliers)
```

```
(array([ 70, 186, 206, 236, 298, 331, 407, 472, 479]), array([0, 0, 0, 0, 0, 0, 0, 0, 0]))
```

7. Feature Engineering

```
## Feature 1: Movie Age
df['movie_age'] = 2026 - df['year_of_release']

## Feature 2: Rating Category
df['rating_category'] = pd.cut(
    df['imdb_rating'],
    bins=[0,5,7,10],
    labels=['Low', 'Medium', 'High']
)

## Feature 3: Number of Actors
df['actor_count'] = (
    df['actors']
    .str.split(',')
    .apply(len)
)

## Feature 4: Story Length
df['story_length'] = (
    df['story']
    .str.len()
)

## Feature 5: Genre Count
df['genre_count'] = (
    df['genres']
    .str.split(',')
    .apply(len)
)
```

8. Final Dataset

```
df.head()
```

	title_x	imdb_id	\
0	Uri: The Surgical Strike	tt8291224	
1	Battalion 609	tt9472208	
2	The Accidental Prime Minister (film)	tt6986710	
3	Why Cheat India	tt8108208	
4	Evening Shadows	tt6028796	

	poster_path	\
0	https://upload.wikimedia.org/wikipedia/en/thum...	
1	Unknown	
2	https://upload.wikimedia.org/wikipedia/en/thum...	

```
3 https://upload.wikimedia.org/wikipedia/en/thum...
4 Unknown
```

```

                                wiki_link \
0 https://en.wikipedia.org/wiki/Uri:_The_Surgica...
1 https://en.wikipedia.org/wiki/Battalion_609
2 https://en.wikipedia.org/wiki/The_Accidental_P...
3 https://en.wikipedia.org/wiki/Why_Cheat_India
4 https://en.wikipedia.org/wiki/Evening_Shadows
```

```

                                title_y                                original_title
is_adult \
0 Uri: The Surgical Strike Uri: The Surgical Strike
0
1 Battalion 609 Battalion 609
0
2 The Accidental Prime Minister The Accidental Prime Minister
0
3 Why Cheat India Why Cheat India
0
4 Evening Shadows Evening Shadows
0
```

```

year_of_release runtime genres ... \
0 2019 138 Action|Drama|War ...
1 2019 131 War ...
2 2019 112 Biography|Drama ...
3 2019 121 Crime|Drama ...
4 2018 102 Drama ...
```

```

                                summary                                tagline \
0 Indian army special forces execute a covert op... No Tagline
1 The story of Battalion 609 revolves around a c... No Tagline
2 Explores Manmohan Singh's tenure as the Prime ... No Tagline
3 The movie focuses on existing malpractices in ... No Tagline
4 Under the 'Evening Shadows' truth often plays... No Tagline
```

```

                                actors
wins_nominations \
0 Vicky Kaushal|Paresh Rawal|Mohit Raina|Yami Ga...
4 wins
1 Vicky Ahuja|Shoaib Ibrahim|Shrikant Kamat|Elen...
0 Wins
2 Anupam Kher|Akshaye Khanna|Aahana Kumra|Atul S...
0 Wins
3 Emraan Hashmi|Shreya Dhanwanthary|Snighdadeep ...
0 Wins
4 Mona Ambegaonkar|Ananth Narayan Mahadevan|Deva... 17 wins & 1
nomination
```

	release_date	movie_age	rating_category	actor_count	\
0	11 January 2019 (USA)	7	High	1	
1	11 January 2019 (India)	7	Low	1	
2	11 January 2019 (USA)	7	Medium	1	
3	18 January 2019 (USA)	7	Medium	1	
4	11 January 2019 (India)	8	High	1	

	story_length	genre_count
0	394	1
1	756	1
2	467	1
3	411	1
4	701	1

[5 rows x 23 columns]